

Ankit Kumar

Machine Learning Engineer | Agentic AI, RAG & Production ML Systems

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Seattle, WA | F-1 STEM OPT through May 2028

MLE with 4+ years of experience building and deploying ML and GenAI systems—from hybrid RAG pipelines with retrieval, reranking, and evidence-gated generation to physics-constrained models running on embedded hardware. Takes extreme ownership from ambiguous requirements through implementation and validation, challenges assumptions with data, and brings technical and business teams together to deliver reliable, user-focused solutions.

EXPERIENCE

Milwaukee Electric Tool Corporation

July 2025 - July 2026

Machine Learning Engineer - Front End Innovation (FEI)

- Led end-to-end development of a physics-informed PyTorch thermal warning system combining a three-body ODE with electrical heat generation; translated requirements from Product Management into model behavior, warning logic, validation criteria, and embedded C specifications, achieving 2.2°C maximum RMSE across 30 held-out profiles and warnings up to 60 seconds before hard shutoffs.
- Cut experiment iteration from 2 hours to 40 minutes by building Azure-based ETL and training pipelines that synchronized lab and field-collected tool-usage data, automated preprocessing and model execution, and used MLflow to track datasets, parameters, metrics, artifacts, and model versions — enabling faster model iteration ahead of milestone deadlines.
- Reduced validation time from 30 days to 3 days by proving equivalent internal heat generation between matched field-use cutting and laboratory load-brake cycles; used ablation and counterfactual analyses from -20°C to 40°C to identify a sensor-induced 4–5°C error, supporting its removal and \$100K in projected hardware savings.
- Drove the PyTorch-to-embedded-C implementation in partnership with the principal firmware engineer, building a time-causal Python reference twin and Azure-hosted Streamlit app for real-time telemetry comparison and session replay; identified two release-blocking OCV/DCIR and initialization defects before they could corrupt field feedback and drive decisions off invalid data, avoiding an estimated 30+ days of firmware and validation rework.

Mitra | Mason Innovation Exchange, GMU

June 2024 - May 2025

Founder & Lead ML Engineer - Selected University Venture

- Owned development of an AI-powered university decision-support platform from problem definition and architecture through evaluation, deployment, and a controlled alpha with 200+ prospective student users, translated ambiguous information needs into requirements for ingestion, retrieval, agent behavior, and evidence-grounded responses while directing student contributors across infrastructure and application development.
- Led the design and implementation of an event-driven AWS data platform (Scrapy, Lambda, SQS, S3, Glue, Athena, PySpark) to process 8K+ university webpages per crawl, combined deterministic rules with GPT-3.5-turbo few-shot classification and Pydantic-validated accept, reject, and retry routing to filter approximately 70% of low-value content, then cleaned, deduplicated, chunked, enriched, and preserved source provenance across a 50K+ chunk corpus through incremental Parquet updates completed in under 30 minutes.
- Improved relevant-context retrieval from 42% to 87% on approximately 50 manually labeled queries by combining 384-dimensional MiniLM SentenceTransformer embeddings, FAISS HNSW candidate search, lexical retrieval, weighted Reciprocal Rank Fusion, and cross-encoder reranking.
- Engineered a bounded LangGraph/LangChain RAG agent for intent classification, canonical entity and conversational-context resolution, and multi-intent decomposition; constrained execution through Pydantic-validated state, fixed retrieval, reranking, and verification tools, per-tool timeouts, and a five-action budget with evidence-gated generation and explicit clarification, retry, and abstention paths.
- Built a reinforcement-learning feedback loop by training a DistilBERT reward model on answer-level user feedback and using trIX PPO to optimize the ranking policy against the learned reward signal, allowing validated user preferences to influence subsequent retrieval rankings.
- Deployed SageMaker-hosted retrieval and reranking services with a Streamlit decision-support application through CI/CD using automated tests, versioned artifacts, and rollback; instrumented LangSmith and CloudWatch tracing to analyze retrieval failures, tool execution, grounding, latency, and agent behavior.

Accenture

February 2021 - July 2023

Packaged App Development Analyst

- Partnered with quantitative analysts and business stakeholders to translate investment-research needs into requirements for an enterprise financial-analytics platform; built PySpark and SQL pipelines that transformed point-in-time fundamentals across thousands of public companies into partitioned Parquet datasets on Amazon S3.
- Developed pooled LSTM/XGBoost and per-company SARIMA models to forecast company KPIs up to two years forward, achieving 92% directional accuracy during stable market regimes and improving transition-period performance by 35% over the existing forecasting baseline.
- Fine-tuned FinBERT on earnings calls, market news, and analyst reports using forward price movement as the supervision signal; integrated the resulting market-impact score with structured financial features to help institutional analysts identify company-growth, valuation, and competitor risks not captured by quarterly fundamentals.

PROJECT

PyLinguist: Cross-Lingual Python Translation

November 2024 – Present

Research project | Submitted to ACL BEA

- Built a two-stage Python translation system across seven languages using token-aware translation, a curated dictionary of 234 Python terms, and Google Translate followed by GPT-4o or Claude 3.5 Haiku refinement, preserving code syntax and structure across diverse writing systems.
- Developed a back-translation evaluation framework combining BLEU, syntax validity, structural preservation, token matching, semantic similarity, and bilingual human review; across 10 held-out programs per language, GPT-4o achieved a 92.5% average overall score versus 61.5% for the baseline and maintained 100% syntax validity, while identifying that additional few-shot examples degraded Claude's performance in some languages without affecting GPT-4o's.

TECHNICAL SKILLS

Languages & Data: Python, SQL, C, PySpark, ETL, Parquet

GenAI & Agents: RAG, LangGraph, LangChain, Pydantic, Prompt Engineering, Evidence-Gated Generation

Retrieval & Preference Learning: SentenceTransformers, FAISS HNSW, Hybrid Search, Weighted RRF, Cross-Encoder Reranking, DistilBERT, PPO, trIX

ML & Evaluation: PyTorch, XGBoost, LSTM, SARIMA, FinBERT, Time-Series Forecasting, LLM/Agent Evaluation, Model Validation, BLEU

Cloud & MLOps: AWS (SageMaker, Lambda, S3, SQS, Glue, Athena, CloudWatch), Azure, MLflow, LangSmith, CI/CD, Automated Testing, Streamlit

EDUCATION

George Mason University

August 2023 - May 2025

M.S. in Computer Science, Machine Learning Concentration | GPA: 3.90/4.00

Honors: Distinguished Academic Achievement Award

CERTIFICATIONS

- AWS Certified Machine Learning Engineer - Associate (MLA-C01)
- AWS Certified Developer - Associate (DVA-C02)